TS ACADEMY

File Handling & NumPy Arrays

Learn to read, write, and manage files in Python — then supercharge your data work with NumPy.

PART 1 · FILE HANDLING

What Is File Handling?

File handling gives your program the ability to work with files on your computer — just like how you use Notepad or Word to open, read, and write documents.

📊
CSV Files
Datasets

🔗
JSON Files
APIs & Web

📝
Text Files
Logs & Reports

💾
Model Output
Results & Exports

ANALOGY

Think of It Like a Diary

📖

Open It

Access the file to read or write

1

✏️

Read or Write

Process the data inside

2

🔒

Close It

Release so others can use it

3

BASICS

Opening & Closing Files

Use Python's built-in `open()` function:

```python
# Opening a file file = open('data.txt', 'r') # 'r' = read mode content = file.read() # Read the entire file print(content) file.close() # Always close!
```

FILE MODES

Three Essential Modes

MODE	MEANING	BEHAVIOR
`"r"`	**Read**	Opens for reading — file must exist
`"w"`	**Write**	Creates or *overwrites* the file
`"a"`	**Append**	Adds to end — keeps existing content

WRITE

```
file = open("data.txt", "w") file.write("Hello World") file.close()
```

APPEND

```
file = open("data.txt", "a") file.write("\nNew line") file.close()
```

BEST PRACTICE

The `with` Statement

A context manager that **automatically closes** the file — even if errors occur.

✗ MANUAL WAY

```
file = open('data.txt', 'r') print(file.read()) file.close() # Easy to forget!
```

✓ THE SMART WAY

```
with open('data.txt', 'r') as file: print(file.read()) # Auto-closed here!
```

WARNING

What Happens If You Don't Close?

💧

Memory Leaks

Wasted memory your program never releases

🔐

File Locks

Other programs can't access the file

🚫

System Limits

OS limits how many files can be open

⚠️

Corruption Risk

Data might not be fully written to disk

PART 2

Introduction to NumPy Arrays

The backbone of data science in Python — fast, efficient, and the foundation for Pandas, Scikit-learn, and TensorFlow.

WHY NUMPY?

Lists vs. NumPy

✗ PYTHON LISTS

```
numbers = [1, 2, 3, 4] new = [] for n in numbers: new.append(n + 5) print(new) # [6, 7, 8, 9]
```

✓ NUMPY ARRAYS

```
import numpy as np arr = np.array([1, 2, 3, 4]) print(arr + 5) # [6 7 8 9]
```

50–100× faster Less memory No loops needed

BENEFITS

Why Data Scientists Love NumPy

⚡
Speed
50-100x faster than Python lists

📦
Memory
Uses significantly less memory

🔢
Vectorization
Operate on entire arrays at once

🧮
Functions
Thousands of math operations

🏗️
Foundation
Powers Pandas, Scikit-learn, TF

🖼️
Real-World
Images, finance, genomics

CREATING ARRAYS

1D & 2D Arrays

1D Array — A single line of values

```
import numpy as np arr = np.array([10, 20, 30, 40]) print(arr) # [10 20 30 40]
```

2D Array — Rows & columns (matrix)

```
arr_2d = np.array([ [1, 2, 3], [4, 5, 6] ]) print(arr_2d) # [[1 2 3] # [4 5 6]]
```

ATTRIBUTES

Array Properties

Every NumPy array carries metadata you can inspect:

```python
arr = np.array([[1, 2, 3], [4, 5, 6]]) print(arr.shape) # (2, 3) → 2 rows, 3 columns
print(arr.ndim) # 2 → two dimensions print(arr.dtype) # int64 → data type
```

`.shape`	`.ndim`	`.dtype`
Rows × Columns	Number of dimensions	Data type (int, float…)

OPERATIONS

Math & Reshaping — No Loops

Vectorized Math

```
arr = np.array([1, 2, 3, 4]) print(arr + 10) # [11 12 13 14] print(arr * 2) # [ 2 4 6 8] print(arr ** 2) # [ 1 4 9 16]
```

Reshaping

```
arr = np.array([1,2,3,4,5,6]) reshaped = arr.reshape(2, 3) print(reshaped) # [[1 2 3] # [4 5 6]]
```

INDEXING & SLICING

Accessing Elements

1D Indexing

```
arr = np.array([10,20,30,40,50]) print(arr[0]) # 10 (first) print(arr[-1]) # 50 (last)
print(arr[1:4]) # [20 30 40]
```

2D Indexing

```
m = np.array([[1,2,3], [4,5,6]]) print(m[0, 1]) # 2 (row 0, col 1) print(m[1, 2]) # 6 (row 1,
col 2)
```

Filtering

```
arr = np.array([1, 2, 3, 4, 5]) print(arr[arr > 2]) # [3 4 5] — Boolean mask filtering
```

WHAT'S NEXT

Keep Learning

Video Tutorials

- Dave Gray — Python File Handling
- NumPy Tutorial Playlist — Data Science Entry

Documentation

- docs.python.org — Official Python Docs
- w3schools.com/python — Tutorials & Examples

Practice with the Jupyter notebook & ask questions in the group!